INSTITUTE CO{)ELINE

by Rihal

AI for Office Automation

DAY 3 — TASKS WORKBOOK

Session Duration: 4 Hours | Hands-On Practice

This workbook contains 10 structured tasks covering AI-powered data analysis, insight extraction, mini-automation building, no-code tool design, and a Capstone Workflow Lab. Tasks are designed for both general business users and software developers — with dedicated exercises for each track.

Workbook Structure

Section 1	AI + Data Analysis	Tasks 1–3	~60 min
Section 2	Insight Extraction & Reports	Tasks 4–5	~45 min
Section 3	Building Mini-Automations	Tasks 6–7	~60 min
Section 4	No-Code Automation Design	Task 8	~30 min
Section 5	Capstone Workflow Lab	Tasks 9–10	~45 min

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■ How to Use This Workbook

Symbol	Meaning
■ Blue box	General business track — relevant to all participants
■■■ Purple box	Developer track — additional exercises for software developers
■ Yellow box	Tip, warning, or important guidance
■ Green box	Sample data provided — use this for the exercise
[blank box]	Space for your written answer or prompt
■ Time	Suggested time allocation for the task
● Difficulty	Foundation / Intermediate / Advanced / Capstone

■ Before You Start

You will need access to an AI tool for all tasks in this workbook. Recommended: **ChatGPT-4o** or **Claude** — both support file uploads. For Tasks 1–5, you will use the sample datasets provided in the workbook. For Tasks 6–10, you will design workflows for your own work context. All tasks can be completed without uploading real company data.

SECTION 1 | AI + Data Analysis

In this section you will practice uploading and interpreting data using AI tools. You will use the sample Sales Dataset provided below for Tasks 1–3.

■ Sample Dataset — Weekly Sales Data (Use for Tasks 1, 2 & 3)

Copy and paste the following CSV data into your AI tool, or type 'use the sales dataset I provided earlier' in follow-up prompts.

```
order_id,date,region,product,category,qty,unit_price,revenue,rep_name,status
1001,2024-03-01,North,Widget A,Hardware,12,45.00,540.00,sara ali,Closed
1002,2024-03-01,South,Widget B,Software,5,120.00,600.00,AHMED NASSER,Closed
1003,03/02/2024,North,Gadget X,Hardware,8,89.99,719.92,Sara Ali,Closed
1004,2024-03-03,East,Widget A,Hardware,,,0.00,Khalid Omar,Pending
1005,2024-03-04,West,Service Pack,Services,2,250.00,500.00,nour hassan,Closed
1006,2024-03-05,South,Gadget X,Hardware,15,89.99,1349.85,Ahmed Nasser,Closed
1007,2024-03-05,North,Widget B,Software,3,120.00,360.00,sara ali,Closed
1008,2024-03-06,East,Service Pack,Services,1,250.00,250.00,KHALID OMAR,Closed
1009,2024-03-07,West,Widget A,Hardware,20,45.00,900.00,Nour Hassan,Closed
1010,2024-03-08,North,Gadget X,Hardware,6,89.99,N/A,Sara Ali,pending
1011,2024-03-08,South,Widget B,Software,8,120.00,960.00,Ahmed Nasser,Closed
1012,2024-03-09,East,Widget A,Hardware,4,45.00,180.00,khalid omar,Closed
1013,2024-03-10,West,Gadget X,Hardware,11,89.99,989.89,Nour Hassan,Closed
1014,2024-03-11,North,Service Pack,Services,3,250.00,750.00,Sara Ali,Closed
1015,2024-03-11,South,Widget A,Hardware,7,45.00,315.00,ahmed nasser,Closed
1016,2024-03-12,East,Widget B,Software,2,120.00,240.00,Khalid Omar,Closed
1017,2024-03-13,West,Service Pack,Services,5,250.00,1250.00,Nour Hassan,Closed
1018,2024-03-14,North,Gadget X,Hardware,9,89.99,809.91,Sara Ali,Closed
1019,2024-03-15,South,Widget A,Hardware,99,45.00,4455.00,Ahmed Nasser,Closed
1020,2024-03-15,East,Service Pack,Services,,250.00,0.00,Khalid Omar,Pending
```

■ What's wrong with this data? — Spot the Issues Before You Start

This dataset has been deliberately designed with quality issues that AI should catch. As you work through Tasks 1–3, watch for: inconsistent date formats, mixed-case names, missing values, a revenue value of 'N/A', an outlier quantity, and inconsistent status casing. Part of your job is to verify that AI correctly identifies these.

1 Data Upload & Quality Assessment

■ General + ■■■ Developer ● Foundation ■ 20 minutes

OBJECTIVE

Practice uploading a dataset to an AI tool, asking it to describe the structure, and identify data quality issues before any analysis begins.

PART A — Write Your Upload Prompt

Using the CRTFC framework, write a complete prompt that: (1) gives the AI context about the dataset, (2) asks it to describe the structure, (3) asks it to identify all data quality issues it can find.

Write your upload + quality assessment prompt here:

Context: I am a business analyst working with a weekly sales dataset in CSV format.

Role: Act as a senior data analyst.

Task:
1. Describe the dataset structure (rows, columns, column meanings, data types).
2. Identify ALL data quality issues present in the dataset.

Format:
- Section 1: Dataset overview (rows, columns, column descriptions)
- Section 2: Data quality issues (bullet points)

Constraints:
- Be precise and complete
- Do not assume missing data
- Flag inconsistencies clearly

PART B — AI Output Analysis

After running your prompt, answer these questions:

Question	Your Answer
How many rows and columns did AI identify?	Rows: 20 Columns: 10
List ALL data quality issues AI found:	• Inconsistent date formats (YYYY-MM-DD vs MM/DD/YYYY) • Mixed casing in rep_name (e.g., “sara ali”, “Sara Ali”, “AHMED NASSER”) • Mixed casing in status (“Closed”, “pending”, “Pending”) • Missing values (qty, revenue) • Revenue contains non-numeric value (“N/A”) • Outlier: qty = 99 • Blank fields in some rows • Inconsistent formatting in names
Which issues did AI miss (if any)?	Logical inconsistency (revenue doesn’t always match qty × unit_price) Duplicate-like names due to casing
Did AI correctly describe what the data represents?	Yes — sales transactions dataset

PART C — Developer Extension

■■■ Developer Task — Write a Data Validation Script

Ask the AI to generate a **Python pandas script** that programmatically validates the dataset. The script should: (1) Load the CSV. (2) Check for missing values per column. (3) Detect inconsistent casing in the 'rep_name' and 'status' columns. (4) Identify rows where 'revenue' is non-numeric. (5) Print a validation report.

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■■■ Write your Python validation script request prompt:

Act as a senior Python data engineer.

Write a pandas script that:
1. Loads the CSV dataset
2. Checks missing values per column
3. Detects inconsistent casing in 'rep_name' and 'status'
4. Identifies non-numeric values in 'revenue'
5. Prints a clear validation report

Output clean, runnable Python code only.

After running the prompt, paste the key output or summarize what the script does:

Script output summary:

- Shows missing values count
  - Lists inconsistent name/status formats
  - Flags “N/A” revenue
  - Prints validation report

2 Data Cleaning via AI

■ General + ■■■ Developer ● Intermediate ■ 20 minutes

OBJECTIVE

Use AI to clean the sales dataset: fix inconsistent casing, standardize date formats, handle missing values, and flag the outlier. Then compare the cleaned result.

PART A — The Cleaning Prompt Chain

Write a **two-turn cleaning workflow**. Turn 1 asks AI to list all cleaning actions it will take. Turn 2 instructs AI to perform all cleaning and output the cleaned CSV.

Turn 1 — Cleaning Plan prompt:

Act as a senior data analyst.

Review this dataset and list ALL cleaning steps needed.

Include:
- casing fixes
- date standardization
- missing values handling
- invalid values
- outliers

Output as a numbered list.

Turn 2 — Execute Cleaning prompt:

Now perform all the cleaning steps.

Requirements:

- rep_name → Title Case
- status → only "Closed" or "Pending"
- dates → YYYY-MM-DD
- revenue "N/A" → NaN
- fill missing qty/revenue with 0 and flag row
- flag qty > 50 as outlier

Output the cleaned dataset as CSV

PART B — Cleaning Verification

After cleaning, verify the AI's work by answering these questions:

Cleaning Item	Expected Result	AI Result ✓/✗	Notes
rep_name casing	All Title Case	✓	All fixed
status casing	All 'Closed' / 'Pending'	✓	Standardized
Date formats	All YYYY-MM-DD	✓	fixed
Missing qty/revenue	Flagged or filled	✓	Done
Revenue 'N/A'	Converted or flagged	✓	NaN
Outlier qty (99)	Flagged as outlier	✓	Not removed

■ Reflection — The Outlier Decision

Order 1019 has qty = 99 — nearly 5x the next highest order. AI should flag this, but should it remove it? In the box below, write your decision and reasoning:

Your outlier decision and reasoning:

Do not remove the outlier automatically. It may represent a real high-value sale. Instead, flag it for business review before making a decision.

PART C — Developer Extension

■■■ Write a Pandas Cleaning Script

Ask AI to generate a complete Python pandas cleaning script for this dataset that: (1) Applies Title Case to rep_name and Proper Case to status. (2) Parses all dates to YYYY-MM-DD. (3) Converts revenue 'N/A' to NaN. (4) Fills missing qty/revenue with 0 and flags the row with a new column 'data_issue=True'. (5) Flags rows where qty > 50 as potential outliers. (6) Saves cleaned output to sales_clean.csv.

■■■ Developer cleaning script prompt:

```
Act as a Python data engineer.

Write a pandas script that:
1. Converts rep_name to Title Case
2. Standardizes status values
3. Converts dates to YYYY-MM-DD
4. Converts 'N/A' in revenue to NaN
5. Fills missing qty/revenue with 0
6. Adds column 'data_issue=True' for affected rows
7. Flags qty > 50 as outliers
8. Saves cleaned file as sales_clean.csv

Return only runnable code.
```

3 Data Analysis & Insight Extraction

■ General + ■■■ Developer ● Intermediate ■ 20 minutes

OBJECTIVE

Using the (cleaned) sales dataset, apply the 6-step Insight Extraction workflow to produce a ranked analysis and 5 actionable business insights.

PART A — Run the 6-Step Workflow

Execute each step as a separate prompt turn. Record the key output of each step.

Step	Your Prompt (summarized)	Key Output from AI
1 — Describe	Describe dataset.	Sales by region/product
2 — Summarize	Summarize key metrics.	Total revenue, avg
3 — Rank	Rank regions/products.	Top: South/West
4 — Trend	Analyze over time.	Increasing trend
5 — Anomaly	Find anomalies	Qty 99 flagged
6 — Insight	Give insights.	Business actions

PART B — Your Top 5 Insights

Based on AI's analysis, write your final 5 insights in the table below. Use your own words — do not just copy the AI output:

#	Insight (your words)	Supporting data point	Recommended action
1	South region highest revenue.	Multiple high-value orders	Focus sales there

2	Gadget X strong performer.	High revenue entries.	Promote product
3	Data quality issues affect accuracy.	Missing + N/A values.	Improve data entry
4	Large order (99 qty) impacts results.	Outlier detected.	Verify before reporting
5	Services generate consistent revenue.	Stable orders.	Expand service offering

PART C — Developer Extension

■■■ Generate a Python Analysis Script

Ask AI to write a pandas + matplotlib script that produces: (1) A bar chart of total revenue by region. (2) A line chart of daily revenue over the date range. (3) A summary statistics table exported to analysis_report.xlsx. The script should run end-to-end without errors on the cleaned CSV.

■■■ Analysis script request prompt:

Act as a Python data analyst.

Write a pandas + matplotlib script that:
1. Loads cleaned dataset
2. Creates bar chart of revenue by region
3. Creates line chart of daily revenue
4. Generates summary statistics
5. Exports results to analysis_report.xlsx

Ensure code runs without errors.

SECTION 2 | Insight Extraction & Report Generation

4 Customer Feedback Analysis

■ General + ■■■ Developer ● Intermediate ■ 25 minutes

OBJECTIVE

Analyze the customer feedback dataset using AI: categorize entries, assess sentiment, identify top issues, and produce a structured management report.

■ Sample Dataset — Customer Feedback (15 entries)

Copy and paste this into your AI tool:

```
ID,Source,Date,Feedback Text
F001,Email,2024-03-01,"The onboarding process was smooth and the team was very helpful. Loved the welcome kit!"
F002,Survey,2024-03-02,"Delivery took 3 weeks instead of the promised 5 days. Very disappointed."
F003,Survey,2024-03-03,"Product quality is excellent but the packaging was damaged when it arrived."
F004,Support,2024-03-04,"I have been waiting 2 weeks for a refund. Nobody is responding to my emails."
F005,Email,2024-03-05,"Great experience overall. The mobile app is intuitive and easy to navigate."
F006,Survey,2024-03-06,"Prices are competitive but I wish there were more payment options available."
F007,Support,2024-03-07,"The product stopped working after 10 days. Replacement process is complicated."
F008,Email,2024-03-08,"Customer service representative was rude and unhelpful when I called."
F009,Survey,2024-03-09,"Fast shipping and the product is exactly as described. Will definitely reorder."
F010,Support,2024-03-10,"Cannot log into my account. Reset password link does not work. Urgent please."
F011,Email,2024-03-11,"The new product update is amazing. Performance has improved significantly."
F012,Survey,2024-03-12,"Instructions manual is confusing. Took 2 hours to set up a simple device."
F013,Support,2024-03-13,"Website keeps crashing on mobile. I cannot complete my purchase."
F014,Email,2024-03-14,"Best customer support I have ever experienced. Issue resolved in 10 minutes!"
F015,Survey,2024-03-15,"Good product but overpriced compared to competitors offering similar features."
```

PART A — Categorization & Sentiment Prompt

Write a prompt that asks AI to produce the per-entry categorization table (ID | Category | Sentiment | Key Theme | Priority) from the Feedback Categorization Template.

Your categorization prompt:

Context: I am analyzing customer feedback data to understand issues and sentiment.

Role: Act as a senior customer experience analyst.

Task: For each feedback entry, classify into:
- Category (Product Quality, Delivery/Shipping, Customer Service, Technical/App, Pricing/Value, Other)
- Sentiment (Positive, Negative, Neutral)
- Key Theme (short phrase)
- Priority (High, Medium, Low)

Format:
Table with columns:
ID I Category I Sentiment I Key Theme I Priority

Constraints:
- Be consistent in categorization
- Keep themes short (3–5 words)
- Assign High priority to urgent/critical issues only

PART B — Record AI's Categorization Output

Fill in the summary that AI produces (or your own from reviewing the output):

Category	Count	% of Total	Dominant Sentiment
Product Quality	3	20%	Mixed
Delivery / Shipping	1	7%	Negative
Customer Service	4	27%	Negative
Technical / App	3	20%	Negative
Pricing / Value	2	13%	Neutral
Other	2	13%	Positive
TOTAL	**15**	**100%**	Mixed

PART C — Escalation Prompt

Now write a follow-up prompt that asks AI to: (1) Identify the top 3 urgent issues requiring immediate action. (2) Draft a 3-sentence management alert for each. (3) Recommend 3 concrete process improvements.

Your escalation & recommendations prompt:

Act as a senior operations manager.

Based on the categorized feedback:

1. Identify the top 3 urgent issues requiring immediate action
2. Write a 3-sentence management alert for each issue
3. Recommend 3 concrete process improvements

Constraints:
- Focus only on high-impact problems
- Be concise and actionable
- Do not add assumptions beyond the data

Record AI's top 3 recommended improvements:

1. Improve customer service response time with SLA targets
2. Fix login, mobile crashes, and system stability issues urgently
3. Streamline refund and replacement processes with clear timelines

PART D — Developer Extension

■■■ Automated Feedback Classifier

Ask AI to write a Python script that reads feedback.csv, sends each row's text to the OpenAI API (model: gpt-4o-mini) with a classification prompt, and writes the result to feedback_classified.csv. Include: batch processing (10 rows per call), basic error handling, and a final print of sentiment breakdown percentages.

■■■ Automated classifier script prompt:

Act as a Python developer.

Write a script that:
1. Reads feedback.csv
2. Sends feedback text to OpenAI API (gpt-4o-mini)
3. Classifies Category, Sentiment, Theme, Priority
4. Processes in batches of 10 rows
5. Handles API errors safely
6. Saves output to feedback_classified.csv
7. Prints sentiment percentage breakdown

Return clean, runnable Python code only.

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5 Data-to-Report: Full Chain

■ General + ■■■ Developer ● Advanced ■ 20 minutes

OBJECTIVE

Chain a complete 5-turn AI workflow — from raw data to finished executive report — using the sales dataset. Practice multi-turn workflow design with a defined output at each stage.

PART A — Design the 5-Turn Chain

Before writing prompts, design each turn in the table below. Define what goes *in* and what should come *out*:

Turn	Purpose	Input to AI	Expected Output
1	Describe dataset	Raw sales data	Structure + issues
2	Clean data	Raw dataset	Cleaned dataset
3	Analyze data	Cleaned dataset	Metrics + Summaries
4	Generate insight	Analysis results	Key insights
5 — Report	Generate final executive report	Insights from Turn 4	Formatted report, max 300 words, formal tone

PART B — Write the Turn 5 Report Prompt

Using the Report Generator Template from the Explanation Guide, write the full Turn 5 prompt that produces the executive management report from the sales analysis:

Turn 5 — Executive Report prompt (full CRTFC):

Context: I am preparing a management-level report based on analyzed sales data insights.

Role: Act as a senior business analyst.

Task: Write an executive report summarizing key findings, performance insights, and business recommendations.

Format:
- Title
- Executive Summary (start with key finding)
- Key Insights (bullet points)
- Recommendations (bullet points)

Constraints:
- Max 300 words
- Formal and professional tone
- Include only data from insights provided
- Do not invent new data
- Focus on actionable recommendations

PART C — Report Quality Review

After receiving the report from AI, evaluate it against these criteria:

Quality Criterion	Met? (✓/✗/Partial)	Comments / What to fix

Leads with key finding	✓	Clear Summary first
All figures are accurate	✓	Based on dataset
Within word limit	✓	under 300 words
Formal professional tone	✓	Professional
Includes actionable recommendations	✓	Clear actions
No invented data	✓	Fully grounded

SECTION 3 | Building AI Mini-Automations

6 Build Your Sales Summary Automation

■ General + ■■■ Developer ● Intermediate ■ 30 minutes

OBJECTIVE

Design and test a reusable Sales Summary Automation template. You will build the complete [PLACEHOLDER]-based prompt, test it with the sample data, and evaluate the output quality.

PART A — Build the Template

Using the Sales Summary Automation Template from the Explanation Guide as your starting point, adapt it for your own work context. Replace generic labels with your actual department, region names, and relevant metrics. Fill in every [PLACEHOLDER]:

Element	Template Text with [PLACEHOLDERS]
Role	Act as a Senior Sales Analyst for the Middle East region
Task	Write a weekly sales performance summary report for the period: 7/7/2026
Data	[PASTE DATA HERE] 7/7/2026
Section 1	HEADLINE: Overall sales performance summary highlighting growth/decline and key takeaway
Section 2	KEY METRICS TABLE: Total Revenue | Total Orders | Avg Order Value | Top Region
Section 3	TOP PERFORMERS: Top 2 regions and top 2 products by revenue
Section 4	ISSUES / RISKS: Data issues, low-performing regions, anomalies, operational risks
Section 5	RECOMMENDATIONS: 3–5 actionable business recommendations
Format	Max 200 words. Tone: Professional, concise, management-level
Constraints	Do not: Invent data — Exceed word limit — Ignore anomalies use informal language

PART B — Test Your Template

Run your completed template with the sales dataset. Then answer the following evaluation questions:

Question	Your Answer
Did the output follow the exact structure you defined?	Yes — all sections clearly present
Were all figures accurate relative to the data?	Yes — aligned with dataset

Was the tone appropriate for the intended audience?	Yes — professional and clear
What would you change in the template for next time?	% growth comparison Visual indicators ↑↓
Estimated time saved vs writing this manually:	30–45 minutes per report

PART C — Developer Extension

■■■ Make Your Template Production-Ready

Ask AI to: (1) Convert your prompt template into a Python function called generate_sales_report(csv_path, week_date, region) that reads a CSV, calls the OpenAI API with your template, and returns the formatted report as a string. (2) Add a docstring explaining parameters and expected output. (3) Write one unit test that mocks the API call and validates the output structure.

■■■ Python template function request prompt:

Act as a Python developer.

Convert this sales summary prompt into a function:

generate_sales_report(csv_path, week_date, region)

Requirements:
1. Read CSV using pandas
2. Insert data into prompt template
3. Call OpenAI API to generate report
4. Return report as string

Also:
- Add full docstring (parameters + output)
- Write one unit test using mocking to validate structure

Return clean, runnable Python code only.

7 Build a Task Planning Workflow

■ General + ■■■ Developer ● Advanced ■ 30 minutes

OBJECTIVE

Design a complete AI task planning automation using a real or realistic project scenario. You will write the prompt, run it, and produce a structured prioritized task plan.

PART A — Define Your Scenario

Choose one of the scenarios below or describe your own real project context:

Scenario	Description	Select ✓
A — Office Relocation	Your team of 12 must move offices in 6 weeks. Tasks include IT setup, supplier notifications, HR updates, and communication plan.	
B — Product Launch	Launch a new service in 30 days: marketing brief, website update, sales training, customer communication, and monitoring plan.	
C — System Migration	Migrate from old CRM to new platform: data export, mapping, testing, staff training, and go-live cutover plan.	
D — Your Own Project	Describe your own real upcoming project in the box below.	

If Scenario D — describe your project context:

Scenario B - Product Launch

PART B — Write the Task Planning Prompt

Using the Task Planning Automation Template, write your complete prompt for the selected scenario:

Your task planning prompt (full CRTFC with your scenario):

Context: We are launching a new digital service in 30 days.

Role: Act as a senior project manager.

Task:
Create a complete task plan including:
- Task breakdown
- Priorities (High/Medium/Low)
- Dependencies
- Timeline (30 days)
- Risks

Format:
- Table: Task | Owner | Priority | Dependency | Deadline
- Section: Critical Path
- Section: Top 3 Risks

Constraints:
- 10–15 tasks only
- Clear and actionable
- No vague tasks
- Realistic deadlines

PART C — Evaluate the Output

Record the key outputs from AI's task plan:

Output Element	AI's Output (summarized)	Quality (1–5)
Total tasks identified	12 tasks	5
High priority tasks (count)	5 tasks	5
Critical path identified?	yes	5
Top 3 risks listed	Identified	5
Sprint allocation logical?	Logical	4
Missing info flagged?	Partial	4

PART D — Developer Extension

■■■ Engineering Sprint Planning

Use the developer Task Planning template to plan a 2-week engineering sprint. Provide 8–10 realistic Jira-style tickets (e.g. 'Implement JWT authentication endpoint', 'Write unit tests for payment service', 'Fix pagination bug on dashboard'). Ask AI to: estimate story points (Fibonacci), identify dependencies, flag unclear tickets, and allocate into a 40-point sprint.

■■■ Engineering sprint planning prompt:

Act as a senior engineering manager.

Plan a 2-week sprint with 8–10 Jira tickets.

Tasks should include:
- Backend features
- Bug fixes
- Testing tasks

For each ticket include:
- Title
- Story points (Fibonacci)
- Dependencies

Also:
- Allocate into a 40-point sprint
- Flag unclear tickets

Output format:
Table + summary

Constraints:
- Realistic engineering tasks
- Total points ≤ 40

SECTION 4 | No-Code Automation Design

8 Design an AI-Powered Automation Workflow

■ General + ■■■ Developer ● Advanced ■ 30 minutes

OBJECTIVE

Design a conceptual automation workflow using one of the no-code tools covered in Day 3 (Make, Zapier, n8n, or Power Automate). You will map the trigger-to-output chain, identify the AI step, and write the prompt template for that step.

PART A — Select Your Workflow Pattern

Choose the workflow pattern you will design, or define your own:

Pattern	Description	Select ✓
Email Triage	Auto-classify incoming emails by urgency, draft reply, send to folder	
Weekly Report	Scheduled: pull data from sheet → AI writes report → email to team	✓
Feedback Router	Form submission → AI classifies sentiment → route to Airtable + Slack alert	
Meeting Brief	Calendar event trigger → AI summarizes context → prep brief to attendees	
Custom / Your Own	Describe a workflow from your own work context	

PART B — Map the Full Automation Chain

Complete the trigger-to-output chain for your selected workflow:

Chain Step	Your Workflow Detail
TRIGGER What starts this workflow?	Every Monday 9 AM
APP / SOURCE Where does the trigger come from?	Google Sheets (Sales Data)
FILTER (optional) Any condition that must be met?	Only run if new data added
AI STEP What does AI do at this stage?	Analyze data + generate weekly report

AI TOOL Which model / service?	OpenAI
OUTPUT What is produced?	Structured sales summary report
DESTINATION APP Where does output go?	Gmail / Outlook
NOTIFICATION Who is alerted?	Sent to Sales Manager + Team
AUTOMATION TOOL Make / Zapier / n8n / Power Automate	Zapier

PART C — Write the AI Step Prompt Template

Write the full CRTFC prompt template that will run inside your AI step. Include [PLACEHOLDERS] for the variable inputs that change each time the workflow runs:

AI Step prompt template (with [PLACEHOLDERS]):

Context: I am generating a weekly sales report for [DEPARTMENT] covering [DATE_RANGE].

Role: Act as a senior sales analyst.

Task:
Analyze the dataset and generate a structured report including:
- Headline summary
- Key metrics
- Top performers
- Risks/issues
- Recommendations

Data:
[PASTE SALES DATA HERE]

Format:
- Section 1: Headline
- Section 2: Metrics table
- Section 3: Top performers
- Section 4: Risks
- Section 5: Recommendations

Constraints:
- Max 200 words
- Professional tone
- No invented data
- Highlight anomalies

PART D — Developer Extension

■■■ n8n Workflow JSON Design

Ask AI to generate a conceptual n8n workflow JSON structure for your chosen pattern. The JSON should define: trigger node type, HTTP request node (for the AI API call), a basic IF node for filtering, and a final action node. Ask AI to also explain which n8n node handles each step and what credentials are needed.

■■■ n8n workflow design prompt:

Act as an automation engineer.

Generate a conceptual n8n workflow JSON that includes:
- Trigger node (schedule)
- Google Sheets node (fetch data)
- IF node (check if new data exists)
- HTTP node (call OpenAI API)
- Email node (send report)

Also explain:
- What each node does
- Required credentials

Return JSON + explanation.

SECTION 5 | Capstone Workflow Lab

The Capstone Lab is your culminating deliverable for the full 3-day programme. Tasks 9 and 10 together form your Capstone submission: Task 9 is the **Design & Build** phase, Task 10 is the **Presentation Preparation** phase.

9 Capstone — Design & Build Your Workflow

■ General + ■■■ Developer ● Capstone ■ 25 minutes

OBJECTIVE

Design and build a complete, end-to-end AI-powered workflow that solves a real problem from your own work context. This is your personal capstone — it should reflect something you will actually use after this programme.

CAPSTONE DESIGN CANVAS

Complete every field. This canvas is the foundation of your Task 10 presentation.

WORKFLOW NAME	Smart Opportunity Alert System (Email → AI Filter → Instant Notification)
PROBLEM STATEMENT (What takes too long / is done manually?)	Important emails (job offers, hiring messages, salary updates, brand opportunities like Sephora or gifts) get buried in inboxes, causing missed opportunities or delayed responses.
CURRENT STATE (How is this done today? Time cost?)	Manually checking Gmail multiple times a day, Scanning emails one by one, Risk of missing keywords or replying late, Time cost: ~5–7 hours/week
TRIGGER (What event starts the workflow?)	New email received in Gmail inbox
INPUT DATA (What goes in? Format? Source?)	* Source: Gmail * Format: Email subject + body text * Example fields: Subject, Sender, Message body, Timestamp

AI PROMPT TEMPLATE (Write your full CRTFC prompt with [PLACEHOLDERS])	Context:
	You are an assistant that filters emails for important opportunities.
	Role:
	Act as a professional email classifier.
	Task:
	Analyze the email and determine if it is relevant based on specific keywords or intent.
	Format:
	Return output in JSON format:
	{
	"relevant": true/false,
	"category": "job / salary / hiring / brand / other",
	"summary": "short 1–2 sentence summary",
	"urgency": "high / medium / low"
	}
	[PLACEHOLDERS]:
	* Email Subject: {{subject}}
	* Email Body: {{body}}

OUTPUT (Format, word/page limit, audience, destination)	If relevant = TRUE: * Send alert via: WhatsApp / Slack / Microsoft Teams * Message format: New Opportunity Alert, Category, Summary, Sender * Audience: You (personal use) * Length: short notification (under 50 words)
AUTOMATION TOOL (Make / Zapier / n8n / Power Automate / Manual trigger)	* Zapier or Make (Integromat) Flow: 1. Gmail → New Email Trigger 2. AI step (OpenAI / built-in AI module) 3. Filter (only relevant = true) 4. Send notification: Slack / Teams (easy), WhatsApp (via Twilio or WhatsApp API)
REVIEW STEP (Who reviews? How long? What do they check?)	* Reviewer: You * Time: 1–2 minutes per alert * Check: Is it actually relevant? - Is urgency correct?- Any false positives?
TIME SAVED (Estimated hrs/week or hrs/month)	Before: 5–7 hours/week After: 1 hour/week Estimated savings: 4–6 hours/week
RESPONSIBLE AI CHECK (Any privacy/accuracy concerns? How are they handled?)	* Privacy: Emails may contain sensitive info → use trusted tools (Zapier, Make) - Avoid storing unnecessary data * Accuracy: AI may misclassify emails - Add keyword backup filter (job, hiring, salary, etc.) * Mitigation: Keep human review step - Regularly refine prompt

LIVE BUILD — Test Your Workflow Now

Run your workflow live during the session. Paste your prompt into your AI tool with real (anonymized) or realistic sample input data. Record the result:

Paste or summarize your AI output here:

```
{
  "relevant": true,
  "category": "job",
  "summary": "Company is offering a new job opportunity with competitive salary and benefits.",
  "urgency": "high"
}
```

The AI analyzes incoming emails and flags relevant ones (job, hiring, salary, Sephora, gifts), then returns a category, short summary, and urgency level.

■ Quality Self-Check Before Task 10

Check all that apply to your workflow:

- ■ The problem is real and recurring — not a one-off task
- ■ The trigger is clearly defined
- ■ The AI step has a full CRTFC prompt template with [PLACEHOLDERS]
- ■ The output format, audience, and destination are specified
- ■ I have tested the prompt and reviewed the output
- ■ I have considered privacy — no real PII in the prompt
- ■ I have estimated the time saved

10 Capstone — Presentation & Peer Feedback

■ General + ■■■ Developer ● Capstone ■ 20 minutes

OBJECTIVE

Prepare and deliver your 5-minute Capstone presentation. You will also give structured feedback to one peer presentation.

PART A — Presentation Slide Planner

Complete this slide-by-slide planner before presenting. This is your speaker guide — you do not need actual slides unless preferred.

Slide	Title	What you will say (key points — max 30 words per slide)	Time
1	The Problem	Important emails get buried in Gmail. I manually check daily, wasting time and risking missed job, hiring, salary, or brand opportunities like Sephora or gifts.	~60s
2	The Workflow	When a new email arrives, automation sends it to AI for analysis. If relevant, it triggers an instant alert via Slack or WhatsApp with a short summary and urgency level.	~90s
3	The Prompt	Show your CRTFC prompt template with [PLACEHOLDERS] Context: Email filtering assistant Role: Professional classifier Task: Identify relevant emails (job, salary, hiring, Sephora, gifts) Format: JSON output Constraints: No hallucination, short summary Input: Subject: {{subject}} Body: {{body}}	~60s
4	Live Demo / Output	Here's an example output. The AI flags the email as relevant, categorizes it as a job opportunity, summarizes it, and assigns high urgency for immediate attention.	~60s
5	Impact	This reduces inbox checking from hours to minutes, saves 4–6 hours weekly, and ensures I never miss important opportunities again.	~30s

PART B — Peer Feedback Form

After watching a peer's presentation, complete this structured feedback form. Feedback should be specific, constructive, and actionable.

Presenter name: _____

Workflow they presented: _____

Feedback Dimension	Rating (1–5)	Specific Comment
Problem was clearly defined		
Workflow logic was sound		

Prompt template was practical & reusable		
Output quality was demonstrated		
Time saved estimate was credible		
One thing I would improve:	—	The workflow could include a clearer validation or human review step to reduce errors.
One thing I will steal for my own work:	—	Using AI to summarize and prioritize notifications instead of just filtering them.

■ My Personal AI Playbook — Day 3 Additions

Add the templates and workflows you built today to your Personal AI Playbook. This becomes your reference document for using AI at work every day.

Task / Workflow	AI Tool	Template / Pattern	How Often	Est. Time Saved
Data upload & quality check	ChatGPT-4o	Upload Context + QA Prompt	Per dataset	30–60 min
Data cleaning	ChatGPT-4o	Cleaning Prompt Chain	Per dataset	1–2 hrs
Insight extraction + report	ChatGPT-4o / Claude	5-Turn Chain Template	Weekly	2–3 hrs/week
Feedback categorization	Claude	Feedback Categorization Template	Monthly	2–4 hrs/month
Sales summary automation	ChatGPT	Sales Summary Template	Weekly	90 min/week
Task planning	Claude	Task Planning Template	Per project	1 hr/project
[Your Capstone Workflow]				
[Your Custom Addition]				

■ Programme Completion Checklist — 3-Day Summary

Day	Skill	Completed ✓
Day 1	Applied CRTFC framework to write structured prompts	
Day 1	Used advanced techniques: Chain of Thought, Few-Shot	
Day 1	Built a reusable prompt template with [PLACEHOLDERS]	
Day 1	Designed a personal AI workflow using the IMTR cycle	
Day 2	Designed a multi-step AI workflow (Input → Process → Output)	
Day 2	Used role-based and constraint prompting for professional output	
Day 2	Produced an executive memo, meeting minutes, or SOP	
Day 2	Applied multi-document analysis and report-to-presentation	
Day 3	Uploaded data and ran the quality assessment workflow	
Day 3	Cleaned a dataset and verified AI's cleaning actions	
Day 3	Completed the 6-step Insight Extraction workflow	
Day 3	Built at least one mini-automation with a reusable template	
Day 3	Designed an automation chain using a no-code tool	
Day 3	Completed and presented the Capstone Workflow Lab	
All Days	Built a Personal AI Playbook with reusable templates	

■ Congratulations on completing the 3-Day AI for Office Automation programme. Your next step is to run your Capstone workflow on real data in the first week back at work.